

IT611 Assignment1

Advanced System Analysis and Design

Weight: 15%

1. Overview

Each student will be assigned to **one topic** from the list below, conduct independent research using the suggested reference material (and additional sources as needed), and prepare a **slide presentation** to deliver in class. The goal is to develop the ability to independently study a specialized topic in systems/software design, synthesize it into a clear technical narrative, and communicate it effectively to peers.

This assignment simulates a real-world scenario: being asked to research an unfamiliar architectural or design topic and brief a technical team on it.

2. Learning Objectives

By completing this assignment, students will be able to:

- Independently research and synthesize technical material from authoritative sources.
- Explain complex system design concepts clearly to a technical audience.
- Compare design/architectural trade-offs and articulate when a given approach is appropriate.
- Practice professional technical presentation skills.

3. Topic List

Students will be given **one** of the following topics. Suggested starting chapters are provided as a foundation — students are expected to go beyond them with additional reading (papers, documentation, case studies, blog posts from reputable sources, etc.).

Note: Topics are designated by the instructor. No two students may present the same topic.

4. Deliverables

1. Slide File (submitted before the presentation date)

- 20–25 slides (excluding title and references slide)
- Must include: title slide, agenda/outline, core content, at least one diagram or visual model, a real-world example or case study, and a references slide

#	Topic	Suggested Starting Material
1	AI Agents: Platforms, Software Design, and Development	<i>An Illustrated Guide to AI Agents</i> — Ch. 1 (Introduction)
2	Designing Machine Learning Systems	<i>Designing Machine Learning Systems</i> (Chip Huyen) — Ch. 1 & 2
3	Selection of 3 Software Architectures (student chooses 3 to compare)	<i>Fundamentals of Software Architecture, 2nd Ed.</i> (Richards & Ford) — Ch. 13, 14, 18
4	Large Language Models	<i>Hands-On Large Language Models</i> (J. Alamar & M. Grootendorst) — Ch. 1 (Partitioning)
5	Microservices Architecture: Patterns, Trade-offs & Migration Strategies	<i>Building Microservices, 2nd Ed.</i> (Sam Newman) — Ch. 1–2
6	API Design & Management (REST, GraphQL, gRPC)	<i>API Design Patterns</i> (JJ Geewax) — Ch. 1–3
7	Domain-Driven Design for Complex System Modeling	<i>Domain-Driven Design Distilled</i> (Vaughn Vernon) — Ch. 1–4

2. In-Class Presentation

- 20–30 minutes presentation + 10 minutes Q&A

3. Reference List

- Students are expected to go beyond the suggested starting chapter(s). They may use different sources — books, peer-reviewed papers, official documentation, or reputable industry publications

5. Content Requirements

Regardless of topic, the presentation should address:

- **What is it / why does it matter?** — Define the concept and its role in modern system design.
- **Core concepts & mechanics** — The key ideas, components, or patterns involved.
- **Trade-offs** — Strengths, limitations, and when this approach is (or isn't) appropriate.
- **Real-world example** — At least one company, system, or case study that illustrates the concept in practice.
- **Comparison / context** — How this topic relates to alternatives or to other topics in the course.

Criterion	Points
Content Accuracy & Depth — correct, well-researched, goes beyond the suggested chapter(s)	4
Clarity & Organization — logical flow, clear explanations, appropriate for a technical peer audience	3
Use of Examples/Case Studies — real-world grounding of concepts	3
Slide Design & Visuals — clear diagrams, readable slides, professional formatting	2
Presentation Delivery — pacing, clarity of speech, engagement, time management	2
Q&A Handling — ability to answer follow-up questions	1
Total	15

6. Grading Rubric (15 points total, mapped to 15%)

7. Submission Guidelines

- Slide file to be submitted (format: PDF or PPTX) via course Blackboard.
- **Deadline: the day before presentation.**
- File naming convention: LastName_FirstName_Topic#.pptx

8. Presentation Schedule

Presentations will take place over during class sessions starting **YYYY**.

9. Academic Integrity

All content must be properly cited. Directly copying slides, diagrams, or text from source material without attribution or paraphrasing constitutes plagiarism and will be handled per the university's academic integrity policy. Students are encouraged to use AI tools for research assistance but must synthesize and present the material in their own understanding — the instructor may ask clarifying questions during Q&A to verify comprehension.